



REGULATIONS 2024
FLEXIBLE CHOICE BASED CREDIT SYSTEM
DEPARTMENT OF CHEMICAL ENGINEERING
B.Tech. Chemical Engineering
CURRICULUM

S.No	Remarks	Category	Course Code R2024	Course Code R2019	Course Title	Prerequisite	Theory Credits	Practical Credits	Total Credits	CGPA / Non - CGPA	Course Type
1	Exempted for LE	HS	SH1106	19EN106	English for Engineers	-	3	1	4	CGPA	TIP (6 Expts)
2		HS	SH1107	19EN107	Technical Writing and Presentation	-	0	2	2	CGPA	Practical (10 Expts)
3		HS	SH9801	NA	Organizational Behavior and Leadership	-	3	0	3	CGPA	Theory
4	Exempted for LE	HS	SH7801	NA	Human Values and Professional Ethics	-	2	0	2	CGPA	Theory
5	Exempted for LE	HS	SH7101	22HS101	தமிழர் மரபு/ Heritage of Tamils	-	1	0	1	CGPA	Theory (Int Only)
6	Exempted for LE	HS	SH7102	22HS102	தமிழரும் தொழில்நுட்பமும்/ Tamils and Technology	-	1	0	1	CGPA	Theory (Int Only)
7	Exempted for LE	BS	SH2201	19MA201	Calculus and Matrix Algebra	-	4	0	4	CGPA	Theory
8		BS	SH2211	19MA211	Statistics and Numerical Methods	-	3	1	4	CGPA	TIP (6 Expts)
9	Exempted for LE	BS	SH2224	19MA224	Transforms and Differential Equations	-	4	0	4	CGPA	Theory
10	Exempted for LE	BS	SH3204	19PH204	Physics for Chemical Engineering	-	3	1	4	CGPA	TIP (6 Expts)
11	Exempted for LE	BS	SH4208	19CY208	Basic Engineering Chemistry	-	3	1	4	CGPA	TIP (6 Expts)
12	Exempted for LE	BS	SH4804	NA	Chemistry for Technologists	-	3	1	4	CGPA	TIP (6 Expts)
13		BS	SH4801	NA	Environmental Sciences and Sustainability	-	2	0	2	CGPA	Theory
14	Exempted for LE	ES	CS3301	19AI301	Python Programming	-	2	1	3	CGPA	TIP (6 Expts)
15	Exempted for LE	ES	CS3801	NA	Artificial Intelligence and Applications	-	1	2	3	CGPA	Theory (Int Only) and Practical (10 Expts)
16		ES	ME0309	19ME309	Engineering Drawing and Visualisation	-	3	1	4	CGPA	TIP (6 Expts)
17	Exempted for LE	ES	ME0310	19ME310	Engineering Practices Lab	-	0	1	1	CGPA	Practical (6 Expts)
18	Exempted for LE	ES	EC2306	19EE306	Basic Electrical, Electronics and Instrumentation Engineering	-	2	1	3	CGPA	TIP (6 Expts)
19	Exempted for LE	ES	CH0301	19CH301	Principles of Chemical Engineering	-	3	0	3	CGPA	Theory
20		ES	CH0302	19CH302	Basics of Material Science and Solid Mechanics	-	3	0	3	CGPA	Theory
21		PC	CH0801	NA	Process Calculations	-	3	0	3	CGPA	Theory
22		PC	CH0401	19CH401	Fluid Mechanics for Chemical Engineers	-	3	1	4	CGPA	TIP (6 Expts)

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23		PC	CH0403	19CH403	Mechanical Operations	-	3	1	4	CGPA	TIP (6 Expts)
24		PC	CH0404	19CH404	Chemical Process Industries	-	3	0	3	CGPA	Theory
25		PC	CH0802	NA	Chemical Engineering Thermodynamics I	-	3	0	3	CGPA	Theory
26		PC	CH0803	NA	Chemical Engineering Thermodynamics II	CH0802	3	0	3	CGPA	Theory
27		PC	CH0405	19CH405	Heat Transfer	-	3	1	4	CGPA	TIP (6 Expts)
28		PC	CH0406	19CH406	Mass Transfer I	-	3	1	4	CGPA	TIP (6 Expts)
29		PC	CH0407	19CH407	Chemical Reaction Engineering I	-	3	1	4	CGPA	TIP (6 Expts)
30		PC	CH0408	19CH408	Mass Transfer II	19CH406 / CH0406	3	1	4	CGPA	TIP (6 Expts)
31		PC	CH0409	19CH409	Chemical Reaction Engineering II	19CH407 / CH0407	4	0	4	CGPA	Theory
32		PC	CH0804	NA	Process Equipment Design and Drawing Laboratory	CH0405 / CH0408	0	2	2	CGPA	Practical (10 Expts)
33		PC	CH0410	19CH410	Transport Phenomena	19CH405, 19CH408 / CH0405,	4	0	4	CGPA	Theory
34		PC	CH0411	19CH411	Process Instrumentation, Dynamics and Control	-	3	1	4	CGPA	TIP (6 Expts)
35		PC	CH0413	19CH413	Project Engineering Economics	19CH404 / CH0404	3	0	3	CGPA	Theory
36		PC	CH0414	19CH414	Safety in Chemical Industries	-	3	0	3	CGPA	Theory
37		PC	CH0415	19CH415	Computational Programming Laboratory	-	0	2	2	CGPA	Practical (10 Expts)
38		PE	CH0528	19CH528	Petroleum Chemistry and Refining Fundamentals	-	3	0	3	CGPA	Theory
39		PE	CH0529	19CH529	Primary Refining Technology	-	3	0	3	CGPA	Theory
40		PE	CH0530	19CH530	Secondary Refining Technology	-	3	0	3	CGPA	Theory
41		PE	CH0531	19CH531	Refinery Advancement and Environmental Regulations	-	3	0	3	CGPA	Theory
42		PE	CH0532	19CH532	Petroleum Equipment Design	-	3	0	3	CGPA	Theory
43		PE	CH0533	19CH533	Petrochemical Technology	-	3	0	3	CGPA	Theory
44		PE	CH0534	19CH534	Design and Analysis of Experiments	-	3	0	3	CGPA	Theory
45		PE	CH0535	19CH535	Optimization of Chemical Processes	-	3	0	3	CGPA	Theory
46		PE	CH0536	19CH536	Chemical Process Simulation and Analysis	-	3	0	3	CGPA	Theory
47		PE	CH0537	19CH537	Pinch Analysis and Heat Exchange Network Design	-	3	0	3	CGPA	Theory
48		PE	CH0538	19CH538	Chemical Process Flow sheeting	-	3	0	3	CGPA	Theory

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49		PE	CH0539	19CH539	Computational Fluid Dynamics	-	3	0	3	CGPA	Theory
50		PE	CH0540	19CH540	Introduction to Biochemistry	-	3	0	3	CGPA	Theory
51		PE	CH0541	19CH541	Bioprocess Technology	-	3	0	3	CGPA	Theory
52		PE	CH0542	19CH542	Fermentation and Bioprocessing	-	3	0	3	CGPA	Theory
53		PE	CH0543	19CH543	Bioseparation and Downstream Processing	-	3	0	3	CGPA	Theory
54		PE	CH0544	19CH544	Enzyme Immobilization Technology	-	3	0	3	CGPA	Theory
55		PE	CH0545	19CH545	Bioreactor Design	-	3	0	3	CGPA	Theory
56		PE	CH0546	19CH546	Plant Design	-	3	0	3	CGPA	Theory
57		PE	CH0547	19CH547	Plant Layout	-	3	0	3	CGPA	Theory
58		PE	CH0548	19CH548	Design Safety	-	3	0	3	CGPA	Theory
59		PE	CH0549	19CH549	Material Selection	-	3	0	3	CGPA	Theory
60		PE	CH0550	19CH550	Statutory Requirements and Customer Care	-	3	0	3	CGPA	Theory
61		PE	CH0551	19CH551	Plant Utilities	-	3	0	3	CGPA	Theory
62		PE	CH0504	19CH504	Air Pollution and Control	-	3	0	3	CGPA	Theory
63		PE	CH0506	19CH506	Wastewater Treatment	-	3	0	3	CGPA	Theory
64		PE	CH0552	19CH552	Solid Waste Management	-	3	0	3	CGPA	Theory
65		PE	CH0553	19CH553	Environmental Social Risk Assessment	-	3	0	3	CGPA	Theory
66		PE	CH0554	19CH554	Process Safety Management	-	3	0	3	CGPA	Theory
67		PE	CH0555	19CH555	Risk and HAZOP Analysis	-	3	0	3	CGPA	Theory
68		PE	CH0522	19CH522	Bioenergy	-	3	0	3	CGPA	Theory
69		PE	CH0523	19CH523	Green Energy	-	3	0	3	CGPA	Theory
70		PE	CH0524	19CH524	Pinch Technology	-	3	0	3	CGPA	Theory
71		PE	CH0525	19CH525	Hydrogen and Fuel Cell Technology	-	3	0	3	CGPA	Theory
72		PE	CH0526	19CH526	Conventional Energy Sources	-	3	0	3	CGPA	Theory
73		PE	CH0527	19CH527	Smart Energy Systems	-	3	0	3	CGPA	Theory
74		OE	XXXXXX	19XXXXX	Open Electives	-	0	0	0	CGPA	Theory

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75		EEC	CH0701	19CH701	Mini Project	-	0	2	2	CGPA	Project (Int Only)
76	Option 1	EEC	CH0702	19CH702	Project Work Phase I	-	0	2	2	CGPA	Project (Int Only)
77	Option 1	EEC	CH0703	19CH703	Project Work Phase II	-	0	6	6	CGPA	Project
78	Option 2	EEC	SH6802	NA	Start Up	-	0	8	8	CGPA	Project
79		EEC	SH6708	19EY708	Career Development Skills	NIL	1	0	1	CGPA	Theory (Int Only)
80		EEC	SH6709	19EY709	Reasoning Ability	NIL	1	0	1	CGPA	Theory (Int Only)
81		EEC	SH6710	19EY710	Quantitative Ability I	NIL	1	0	1	CGPA	Theory (Int Only)
82		EEC	SH6711	19EY711	Quantitative Ability II	SH6710 / 19EY710	1	0	1	CGPA	Theory (Int Only)
83		EEC	SH6712	19EY712	Employment Enrichment Skills	19EY702 / 19EY709 / SH6709	1	0	1	CGPA	Theory (Int Only)
84		EEC	SH6713	19EY713	Placement Readiness Assessment	19EY704 / 19EY711 / SH6711	1	0	1	CGPA	Theory (Int Only)
85		EEC	SH6707	19EY707	Paper Publication	-	0	0	1	NON-CGPA	Experiential Learning (Int Only)
86		MC	SH8801	NA	Constitutionalism and Artificial Intelligence Policies in India	-	1	0	1	NON-CGPA	Theory (Int Only)
87	Any 1	MC	SH8802	NA	Internship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
88	Any 1	MC	SH8803	NA	Entrepreneurship	-	0	0	2	NON-CGPA	Experiential Learning (Int Only)
89	Any 1	MC	SH6803	19MC807	National Service Scheme	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
90	Any 1	MC	SH8807	19MC808	National Sports Organization	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)
91	Any 1	MC	SH8808	19MC809	Youth Red Cross	-	0	0	0	NON-CGPA	Experiential Learning (Int Only)

BATCH: 2024- 2028 Onwards		
Category	Credits to be Earned (Regular)	Credits to be Earned (Lateral Entry)
Humanities and Science	13	5
Basic Science	26	6
Engineering Science	17	7
Professional Core	58	58
Professional Electives	18	18
Open Electives	12	12
Employability Enhancement Courses	16	16
Mandatory Courses	3	3
TOTAL	163	125